

Emergency Maintenance Report —

EMR-2025-002

APEX EMERGENCY GRID SOLUTIONS

24/7 High-Voltage Rapid Response · NAESB-Certified Emergency Contractor
1450 Industrial Parkway, Tacoma, WA 98421 · Dispatch: (253) 555-0177

Field	Value
Document ID	EMR-2025-002
Issued By	Apex Emergency Grid Solutions
Report Date	2025-06-04
Site	NW-PowerPool-WIND-003
Asset	TRA-10561 (Siemens Energy power transformer, 138 kV / 34.5 kV, 75 MVA)
Outage ID	OUT-2025-1405
Cascadia Work Order	WO-2025-05612
Cascadia Authorization	EPA-2024-11 (invoked 2025-05-26 03:11 PDT)
Apex Crew Lead	M. Castellanos, Lead Transformer Specialist
Distribution	Cascadia Operations · Cascadia Compliance · Cascadia AP

1. Scope of Emergency Work

Apex was dispatched at 03:11 PDT 2025-05-26 in response to a vibration_excess trip on TRA-10561. CPS confirmed inability to mobilize a transformer-qualified crew within 4 hours due to overlapping deployments in MISO. EPA-2024-11 was invoked by the on-call duty manager.

Authorized scope:

- Onsite isolation, oil sampling, and acoustic survey of TRA-10561.
- Stabilization and return to service if achievable within emergency window.
- OEM-coordinated assessment to identify root mechanical cause.

Outage duration to partial restoration: **158 hours**. Lost generation per Cascadia SCADA: 2,615.1 MWh.

2. Findings

Onsite arrival 2025-05-26 08:34 PDT.

1. **Vibration spectrum at the LV bushing turret** showed dominant 120 Hz and 360 Hz components at 18.4 mm/s and 11.2 mm/s respectively — well above the ISO 10816-3 zone C/D boundary of 7.1 mm/s for this class.
2. **Dissolved gas analysis (DGA)** of the main tank oil returned acetylene 14 ppm, ethylene 92 ppm, hydrogen 280 ppm. The acetylene reading is indicative of recent arcing; the IEEE C57.104 ratio places the unit in "Condition 3 — investigate immediately."
3. **Acoustic emission survey** localized partial discharge activity to the HV winding outboard end, phase B, suggesting inter-turn insulation degradation.
4. The unit's most recent **oil-spectrometer test** on file was dated 2024-04-11. Per OEM Siemens Energy Transformer Manual SE-TR-2300 §7.4, oil-spectrometer testing on this class is **annual**. The CSM-7 internal procedure (Cascadia) lists this step as quarterly, but the procedure note on the asset history reads "performed annually per OEM guidance" — which is the cadence Apex followed during prior service visits and is consistent with OEM documentation provided to us. (Cascadia engineering may wish to reconcile internal CSM-7 to OEM Siemens Manual SE-TR-2300.)
5. Mechanical assessment indicates winding clamping force has decayed; the 120 Hz signature is consistent with loosened core-laminations exciting the LV winding.

3. Repairs Performed

Stabilization scope only — full repair is outside emergency authorization and is proposed below.

Item	Detail	Hours / Qty
De-energization, oil sampling, DGA dispatch	Lead + 1 technician	18 hrs
Acoustic emission survey, full enclosure	2 technicians, weekend rate	26 hrs
Bushing turret torque check and re-shimming	Phase A/B/C, LV side	22 hrs
	Includes filter pass	34 hrs

Item	Detail	Hours / Qty
Tank vacuum, partial oil change (2,400 gal)		
Temporary winding restraint installation	Mechanical wedging, OEM-approved	28 hrs
Re-energization with stepped voltage protocol	4 attempts, 1 hold for cooldown	19 hrs
Standby — weekend hold for OEM coordination	1.5× multiplier	11 hrs
Total labor		158 hours

Materials: 2,400 gallons mineral oil (Nynas Nytro Libra equivalent); 1 set bushing turret gaskets (Siemens PN 2GA5108); mechanical wedging kit (Apex PN AP-TXW-118).

4. Recommendations

The temporary winding restraint and oil change will keep TRA-10561 in service in a derated configuration (60 MVA continuous, 65 MVA 30-minute). In Apex's professional judgment:

1. The HV winding inter-turn insulation degradation is unlikely to recover. We recommend **full HV and LV winding replacement** within 180 days. Estimated capital scope: \$1.45M (separate proposal AP-Q-2025-129 to follow).
2. CSM-7 oil-spectrometer testing should be confirmed as annual per OEM guidance; the conflicting quarterly cadence in some Cascadia internal documents appears to be a residual from a predecessor procedure and is not supported by Siemens Energy.
3. Vibration monitoring at the LV bushing turret should be elevated from advisory to action-level above 7.1 mm/s consistent with ISO 10816-3.
4. Apex recommends an OEM-supervised factory acceptance test for whichever replacement option is selected.

5. Invoice Summary

Line	Rate	Qty	Extended
Labor — emergency rate, standard hours	\$420.00/hr	122 hrs	\$51,240.00

Line	Rate	Qty	Extended
Labor — emergency rate, weekend 1.5×	\$630.00/hr	36 hrs	\$22,680.00
Subtotal labor			\$73,920.00
Parts — mineral oil, 2,400 gal	\$11.40/gal	—	\$27,360.00
Parts — bushing gaskets, wedging kit, consumables	itemized	—	\$14,800.00
Emergency parts mark-up	35% on parts subtotal	—	\$14,756.00
OEM (Siemens Energy) remote engineering	pass-through + 15%	—	\$46,000.00
DGA laboratory analysis, expedited	flat	—	\$4,200.00
Crane / lift services (subcontract)	pass-through + 12%	—	\$34,800.00
Travel, per-diem (3 technicians × 7 days)	per Apex tariff	—	\$13,650.00
Mobilization premium (EPA <4hr dispatch, off-hours)	flat	1	\$48,000.00
EPA-2024-11 emergency response surcharge	30% on labor	—	\$22,176.00
Documentation, post-energization test report	flat	—	\$6,800.00
Subtotal			\$306,462.00
Risk / availability premium (per Apex MSA §4.3)	—	—	\$82,538.00
Total Invoice			\$389,000.00

Closing

Equipment returned to derated service 2025-06-01 at 17:11 PDT. Apex notes this is the third transformer-class deployment to NW-PowerPool in FY2025. We are prepared to mobilize the winding-replacement scope under separate proposal AP-Q-2025-129.

Signed	Name	Date
Crew Lead	M. Castellanos	2025-06-04

Signed	Name	Date
Apex Engineering Review	D. Pieterston, PE	2025-06-04
Apex Account Manager	J. Beaumont	2025-06-04

Attachments: DGA lab report · acoustic emission survey traces · vibration spectrum plots (6) · photographs of bushing turret (8) · Apex proposal AP-Q-2025-129 (winding replacement).

Apex references: Siemens Energy Manual SE-TR-2300 §7.4 · ISO 10816-3 · IEEE C57.104.

Cascadia references cited in dispatch packet: EPA-2024-11 · CSM-7 · MOC-2025-022.